

ELD COMPLIANCE SYSTEM – PROJECT COMPLETION REPORT

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Frontend URL: <https://frontend-el-dsystem.vercel.app>

Demo Video: Google Drive Video Link

1. General Overview

The ELD Compliance System is a complete, intelligent, and fully responsive web platform designed to help truck drivers, fleet managers, and administrators comply with FMCSA (Federal Motor Carrier Safety Administration) Hours-of-Service (HOS) regulations. The system enables users to manage daily electronic logs, plan trips with automated geocoding and distance estimation, monitor HOS compliance in real time, and generate FMCSA-compliant daily PDF reports. It was developed with a modern technology stack — React for the frontend, Django REST Framework for the backend, and PostgreSQL as the database — ensuring a scalable, secure, and high-performance solution.

2. Email Verification – Local Testing Context

During development, the backend was hosted on Render.com using its free-tier plan. Render's free plan restricts outbound SMTP connections to external mail servers, which prevents email features such as account verification and password reset from operating while hosted on that environment. To validate and demonstrate the email workflows reliably, all email verification and password reset scenarios were tested locally using a configured SMTP service. The codebase and configuration are production-ready; once deployed to a hosting provider that permits outbound SMTP access, the email features operate as intended.

3. Development Inspiration and Research

This project was inspired by official FMCSA documentation, publicly available training materials, and manual driver log templates. We carefully reviewed the provided PDFs and instructional videos to ensure the system's compliance logic aligns with FMCSA standards. The design and workflows implemented reflect a deep understanding of regulatory requirements and driver responsibilities, translated into a practical and reliable digital solution.

4. Time Commitment and Effort

The project required approximately 32 hours of focused development and refinement. This time was devoted to designing a robust architecture, implementing secure backend logic, creating an advanced trip planning algorithm, building a responsive user interface, and

performing thorough testing across multiple user roles and scenarios. The extended development period ensured a reliable, high-quality result that demonstrates technical precision and professional craftsmanship.

5. Video Presentation

The demonstration video provides a comprehensive walkthrough of the platform's features and workflows, including user registration, admin and manager approval, trip planning, ELD log management, and automated PDF report generation. The video was recorded independently and uploaded for review. The recording is available via Google Drive and the Loom account referenced in the project notes.

6. Conclusion and Appreciation

The ELD Compliance System represents a meaningful professional accomplishment. Despite infrastructure limitations on the free hosting tier, the project was completed with care, technical rigor, and fidelity to FMCSA guidance. I am proud of the outcome and confident that this system demonstrates the ability to design, implement, and deliver complex, compliance-driven software solutions. Thank you for taking the time to review this work. If you have any questions or require additional demonstrations, I am available for further walkthroughs and discussion.

Respectfully,
Robert Niyonzima
Full Stack Developer